

- Understand Explainable AI concepts
- Train a simple loan approval model
- Generate LIME explanations
- Generate SHAP explanations
- Compare local vs global explanations
- Create a simple counterfactual explanation

Pandas

NumPy

Scikit-learn

LIME

SHAP

Matplotlib

No saliency maps required

Lime: <https://www.youtube.com/watch?v=ZMB6TwQ6Vuo>

Shap:

<https://www.youtube.com/watch?v=MQ6fFDwjuc0&list=PLqDyyww9y-1SJgMw92x90qPYpHgahDLIK&index=1>

https://youtu.be/L8_sVRhBDLU?si=9-pqN0u8P8ZXHjoG

Matplotlib: refer previous videos shared

Counterfactual Tutorial: https://youtu.be/8EQwpmK49sk?si=ebwo7PQiVpogi_xj

Saliency: No need to do saliency plots!

Scikit-Learn Tutorial: Refer previous videos shared

Use additional resources as needed

- Recommended: Loan Status Prediction Dataset (See E-Dimension for Dataset)
- Inspect features
- Inspect target variable
- Check class distribution

- Logistic Regression or Random Forest
- Train model
- Evaluate accuracy
- Generate predictions

- Select one applicant
- Generate local explanation
- Identify important features

- Generate SHAP Waterfall Plot
- Observe positive and negative contributions

- Generate SHAP Summary Plot
- Observe global feature importance

- Ease of understanding
- Detail provided
- Trustworthiness

- What would need to change?
- Increase income
- Reduce loan amount
- Improve credit history
- Note: Code is not needed for this – just do a manual thing and paste in PPTX

- Applicant
- Regulator
- Auditor
- Data Scientist
- Note: your group mates can play these roles. Divide them up!

- Model chosen
- LIME findings
- SHAP findings
- Counterfactual explanation
- Stakeholder analysis
- Submit notebook + PPT reflection